

Creating Terraform User

The screenshot shows the AWS IAM console interface for creating a new user. The browser address bar indicates the URL: `us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/users/create`. The user is logged in as **PRATIYASA PRITIMAYI SAHU (881081147231)**.

Specify user details

User details

User name
terraform-user

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☐ **Provide user access to the AWS Management Console - optional**
In addition to console access, users with `SigninLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Info If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel **Next**

The bottom screenshot shows the 'Retrieve access keys' step for the user 'terraform-user'. The browser address bar indicates the URL: `us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/users/details/terraform-user/create-access-key`.

Retrieve access keys

Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key | **Secret access key**

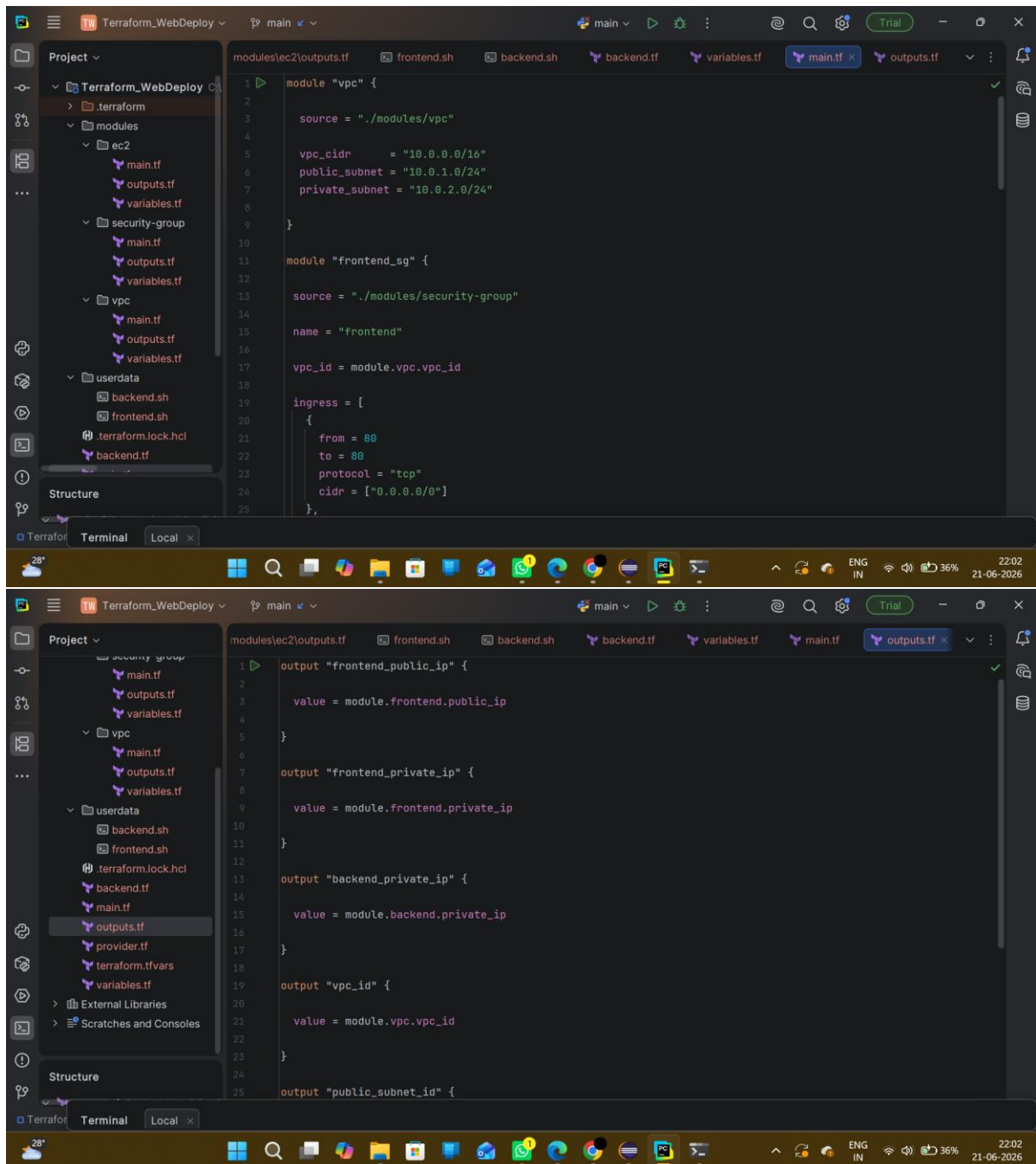
Access key
AKIA42JEHL5PVRBEYVNZ

Secret access key
***** **Show**

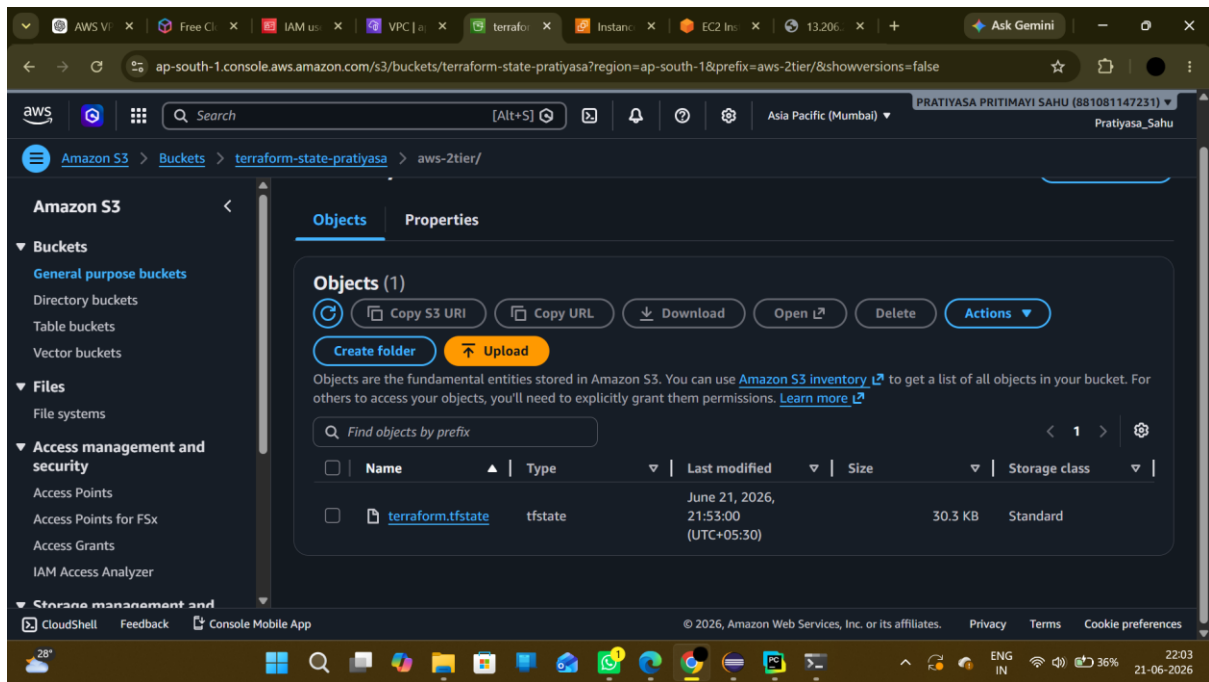
Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.

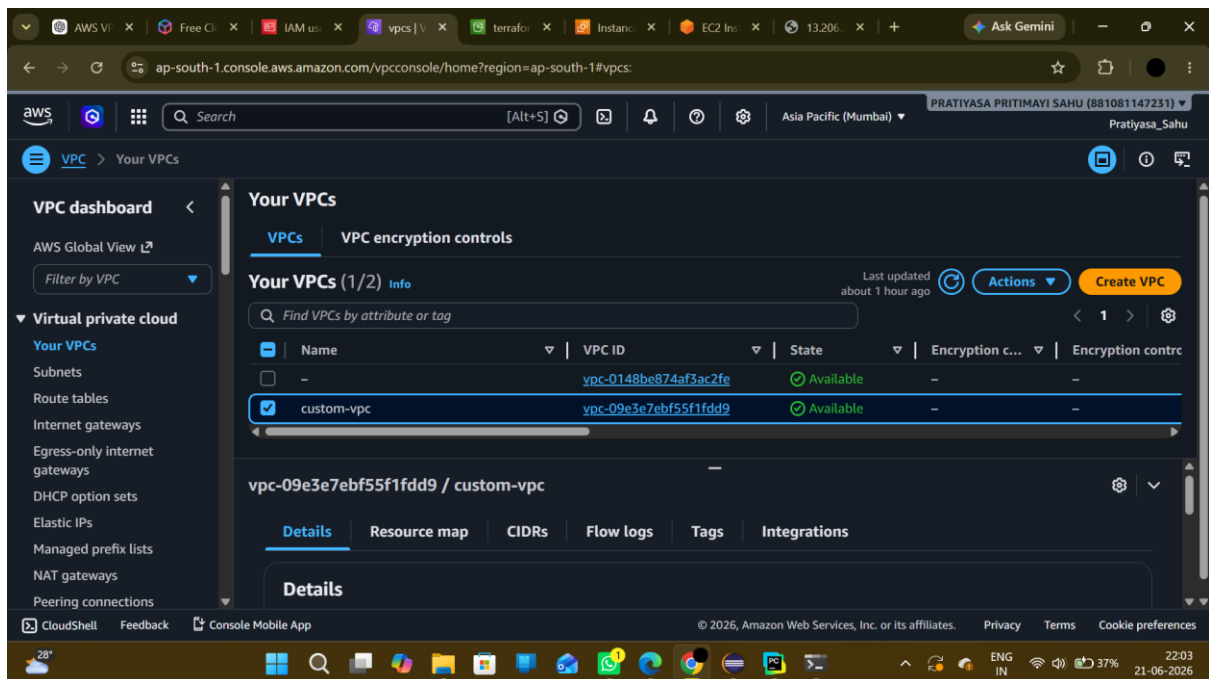
Folder structure for terraform web deploy



Creating S3 bucket for backend.tf file



After terraform apply -> VPC created



Subnet and Route table created

The image displays two screenshots of the AWS Management Console, specifically the VPC dashboard, showing the configuration of subnets and route tables in the ap-south-1 region.

Top Screenshot: Subnets (1/5)

The "Subnets (1/5)" page shows a list of subnets. The "public-subnet" is selected, and its details are displayed below the table.

Name	Subnet ID	State	VPC
-	subnet-06ade738336125d41	Available	vpc-0148be874af3ac2fe
private-subnet	subnet-0730e0fc05cf2ef37	Available	vpc-09e3e7ebf55f1fdd9 custo...
-	subnet-075d148d3c4ad8a7c	Available	vpc-0148be874af3ac2fe
-	subnet-0f656d17349e6b26f	Available	vpc-0148be874af3ac2fe
public-subnet	subnet-0bf682a5df80800d9	Available	vpc-09e3e7ebf55f1fdd9 custo...

Subnet Details: subnet-0bf682a5df80800d9 / public-subnet

The details section shows the "Route table" tab selected, indicating the subnet is associated with a route table.

Bottom Screenshot: Route tables (4)

The "Route tables (4)" page shows a list of route tables. The "public-route-table" is selected, and its details are displayed below the table.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main
-	rtb-02339fc2698c073aa	-	-	Yes
-	rtb-0b0204e37964e97cf	-	-	Yes
private-route-table	rtb-0a054b71b6d954300	subnet-0730e0fc05cf2ef...	-	No
public-route-table	rtb-07184192b7e5ea72c	subnet-0bf682a5df8080...	-	No

Select a route table

Internet gateway and elastic ip created

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'Internet gateways' page for the 'ap-south-1' region. It shows two internet gateways: one with ID 'igw-0331f0c39f836105c' and another named 'custom-igw' with ID 'igw-06ff88ed4d92f62b3'. Both are in an 'Attached' state. The bottom screenshot shows the 'Elastic IP addresses' page, displaying one elastic IP address named 'nat-eip' with the allocated IPv4 address '35.154.186.47' and allocation ID 'eipalloc-07c5ab84173e1543d'. A notification banner at the bottom of the console suggests viewing IP address usage and recommendations.

Internet gateways (2)

Name	Internet gateway ID	State	VPC ID
-	igw-0331f0c39f836105c	Attached	vpc-0148be874af3ac2fe
custom-igw	igw-06ff88ed4d92f62b3	Attached	vpc-09e3e7ebf55f1fdd9 custo

Select an internet gateway above

Elastic IP addresses (1)

Name	Allocated IPv4 addr...	Type	Allocation ID
nat-eip	35.154.186.47	Public IP	eipalloc-07c5ab84173e1543d

Select an elastic IP address

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

Nat Gateway and EC2 instances created

The image shows two screenshots of the AWS Management Console. The top screenshot displays the 'NAT gateways' page in the 'ap-south-1' region. A single NAT gateway, 'nat-gateway' with ID 'nat-0b1a584c7cb563cce', is shown in an 'Available' state. The bottom screenshot shows the 'Instances' page, listing three EC2 instances: 'frontend' (Running), 'backend' (Running), and another 'backend' (Terminated). The selected 'backend' instance (ID: i-08fca42e58d99b0ff) is shown in detail, highlighting that it has 'No Public IP' assigned, with only a private IP of 10.0.2.124.

NAT gateways (1)

Name	NAT gateway ID	Connectivity...	State	State message	Availabi
nat-gateway	nat-0b1a584c7cb563cce	Public	Available	-	Zonal

Instances (1/3)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
frontend	i-0b503ac58cc7673c1	Running	t3.micro	3/3 checks passed	ap-south-1a	-
backend	i-0a4ca6b0c5d577832	Terminated	t3.micro	-	ap-south-1b	-
backend	i-08fca42e58d99b0ff	Running	t3.micro	Initializing	ap-south-1b	-

i-08fca42e58d99b0ff (backend)

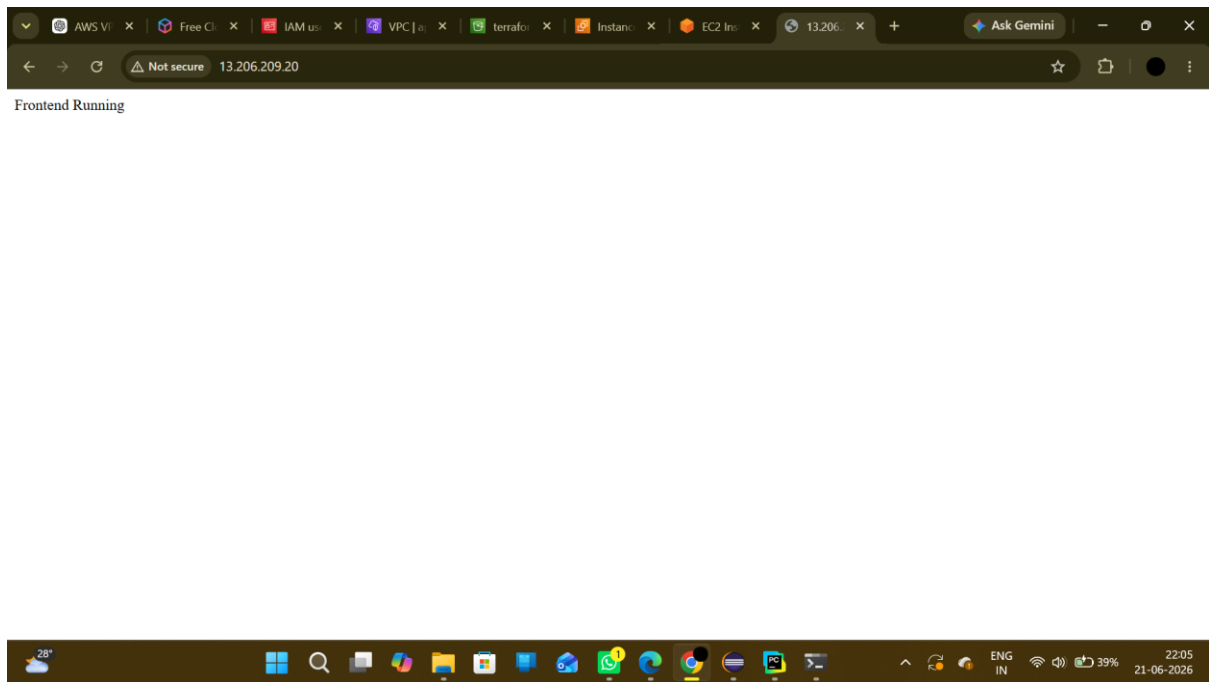
Instance summary

Instance ID: [i-08fca42e58d99b0ff](#)

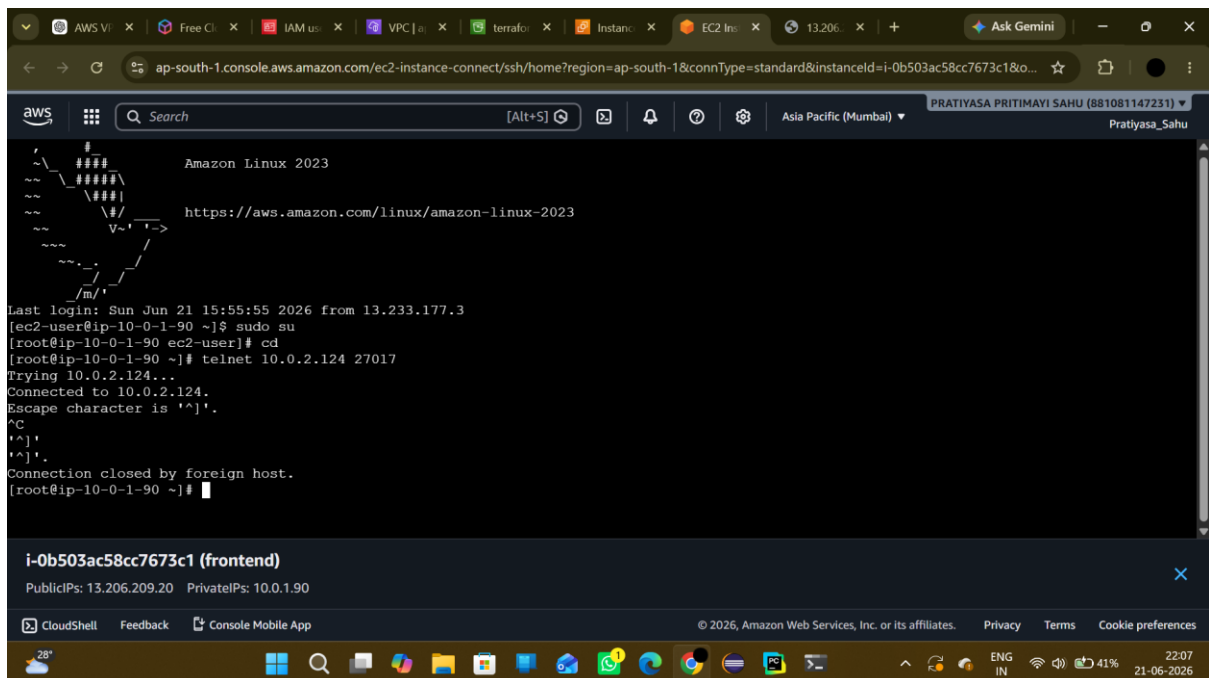
Public IPv4 address: -

Private IPv4 addresses: [10.0.2.124](#)

Webpage can be accessed from browser (Frontend Server)



Backend Server accessed from Frontend Server



Access to connect to backend server is denied

The screenshot shows the AWS Management Console interface for connecting to an EC2 instance. The browser address bar indicates the URL: `ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#ConnectToInstance:instanceId=i-08fca42e58d99b0ff`. The console header shows the user is logged in as PRATIYASA PRITIMAYI SAHU (881081147231) in the Asia Pacific (Mumbai) region.

The navigation pane on the left shows the path: **EC2** > **Instances** > **i-08fca42e58d99b0ff** > **Connect to instance**.

The main content area has four tabs: **EC2 Instance Connect** (selected), **SSM Session Manager**, **SSH client**, and **EC2 serial console**.

Two error messages are displayed in yellow boxes:

- No public IPv4 or IPv6 address assigned**
With no public IPv4 or IPv6 address, you can't use EC2 Instance Connect. Alternatively, you can try connecting using [EC2 Instance Connect Endpoint](#).
- Instance is not in public subnet**
Associated subnet [subnet-0730e0fc05cf2ef37 \(private-subnet\)](#) is not a public subnet.
To use EC2 Instance Connect, your instance must be in a public subnet. [To make the subnet a public subnet, add a route in the subnet route table to an internet gateway.](#)

The **Instance ID** is `i-08fca42e58d99b0ff (backend)`.

The **Connection type** section has two options:

- ☒ **Connect using a Public IP**
Connect using a public IPv4 or IPv6 address
- ☐ **Connect using a Private IP**
Connect using a private IP address and a VPC endpoint

The bottom of the console shows the footer with copyright information: © 2026, Amazon Web Services, Inc. or its affiliates. Links for [Privacy](#), [Terms](#), and [Cookie preferences](#) are provided.

The Windows taskbar at the bottom shows the system clock as 22:07 on 21-06-2026, with a battery level of 41% and network status.